

Title: MLSD Credit Card Fraud Detection
Course: Machine Learning System Design
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Year of Study: 3rd year

Credit Card Fraud Detection MLOps Project

GitHub repository link: <https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops>

1. Introduction

The aim of this project is to create a Credit Card Fraud Detection model that will be able to detect fraudulent transactions by employing Machine Learning algorithms. In addition to developing a fraud detection model, this project showcases implementation of various MLOps practices like experiment tracking, data versioning, containerization, CI/CD pipeline, and deployment on Kubernetes.

Detection of fraud is one of the most critical problems faced by financial institutions as fraudulent transactions lead to substantial financial loss. As cases of fraud occur quite rarely compared to other transactions, there is class imbalance in the dataset, thereby creating a difficult machine learning problem.

In this project, a comprehensive process for machine learning is followed from the stage of data preparation to model development and deployment.

2. Dataset Collection & Understanding

For the purpose of this project, a dataset containing transaction data along with the target feature name is_fraud was used where:

0: Legitimate Transaction

1: Fraudulent Transaction

The dataset was saved locally within the project directory, and Pandas was employed for reading it.

```
df = pd.read_csv("archive/archive/fraudTrain.csv")
```

Once the dataset has been loaded, its shape is checked to get an understanding about how much data is available for model building.

The target column for classification is: **is_fraud**.

3. Data Pre-processing, Model Training, and Testing

Upon loading the fraud detection dataset, thorough pre-processing was done to convert the raw data to data ready to be used in machine learning applications. As most real-world data have issues such as being incomplete, irrelevant, categorical, and having features in different scales, it is crucial to apply the proper pre-processing techniques in order to prepare the data in a manner that can be properly used in the machine learning models. In addition, separating the target class (fraudulent vs. non-fraudulent) from input features, and preparing the data for training were among the other key processes. It is important to standardize and organize the features prior to dividing them into training and testing subsets in order to facilitate learning and increase computation efficiency.

Due to being highly imbalanced, where fraudulent cases are less than 1% of the data records, it was necessary to balance the dataset using appropriate imbalance handling techniques. This step helps the models learn fraud-related patterns in the absence of any bias towards majority classes. In order to select the best-performing models, several machine learning methods were implemented, trained, and finally evaluated.

The performance of the developed models was analyzed by evaluating the quality of classifications based on certain indicators which allow for assessing the efficiency of fraud detection from various angles. Such measures allowed for analyzing the capacity of the model in recognizing transactions involving fraud while minimizing wrong predictions. Additionally, methods were applied to visualize the performance of the models and gain insight into the process of classification performed by each model. According to the obtained outcomes, the optimal model among the tested ones was identified, and further saved for possible usage. Besides the trained model itself, all other necessary components, including preprocessors, were saved to ensure reproducibility and facilitate further integration.

4. Experiment Tracking With MLflow

MLflow is an open source platform. It is used to help to track and manage machine learning experiments. It provides features to log parameters, metrics, and version of the model, as well as the result of the training of the experiment.

For effective machine learning experiment management, MLflow was employed.

MLflow was used to log:

- Name of the model
- ROC-AUC score
- Model Parameters
- Model Artifacts

For example,

```
mlflow.log_param("model", name)
```

```
mlflow.log_metric("roc_auc", auc)
```

MLflow's UI was started using : `mlflow ui`

and can be accessed using : `http://127.0.0.1:5000`

This provided an option for monitoring and comparing several training runs.

5. Dataset Version Control (DVC)

DVC stands for “Data Version Control” and is a specialized version control tool which manages datasets and machine learning models in large projects.

Storing large data files using Git is not advisable. Thus, DVC was utilized for dataset version control.

The training data was version-controlled using : `dvc add archive/archive/fraudTrain.csv`

Advantages:

- Versioning of datasets
- Reproducibility
- Collaboration feature

DVC stores metadata but stores huge files out of the Git repository.

6. Git and GitHub Usage

Git is a distributed version control system which tracks modifications in the source code during the software development process. GitHub is a cloud based platform where we can share the projects, version management in Git repositories.

For version control and collaboration, Git was initialized on the project repository.

Commits were done to keep track of code modifications which are then pushed to GitHub.

Commands used:

```
git add .
```

```
git commit -m "Project update"
```

```
git push origin main
```

GitHub was the repository for source code.

7. Continuous Integration via GitHub Actions

CI/CD stands for Continuous Integration and Continuous Deployment. Automates the process of building, testing and deploying applications.

A CI/CD pipeline was developed through GitHub Actions.

Pipeline file : `.github/workflows/ci.yml`

The pipeline is triggered whenever there is code push to GitHub.

Steps performed:

- Installation of dependencies
- Setting up environment
- Structure validation

Advantages:

- Automation of tests
- Faster development process
- Issue identification at an early stage

8. Docker Containerization

Docker is a containerization platform where packages and its dependencies are stored in a single container.

To provide uniformity in execution irrespective of the system, the project was containerized through Docker.

A Dockerfile was created.

Key steps:

1. Fetch Python base image
2. Define working directory
3. Install dependencies
4. Import project files
5. Run Streamlit app

Creating Docker image : `docker build -t fraud-detection-app .`

Running container : `docker run -p 8501:8501 fraud-detection-app`

Advantages:

- Uniform environment
- Easy to deploy
- Dependency management

9. Kubernetes Deployment

Kubernetes is a container orchestration platform that automatically manages, scales and deploys containerized applications. Kubernetes configuration files were created for simulating the deployment in the production environment. Kubernetes helps in easy deployment.

Files:

1. `k8s/deployment.yaml`
2. `k8s/service.yaml`

Deployment YAML File

Contains:

- Replicas of the application
- Docker image name
- Resource management settings
- Service YAML File

Contains:

- Exposure of application
- Network settings
- Port forwarding settings

Commands for deploying the service:

```
kubectl apply -f k8s/deployment.yaml
```

```
kubectl apply -f k8s/service.yaml
```

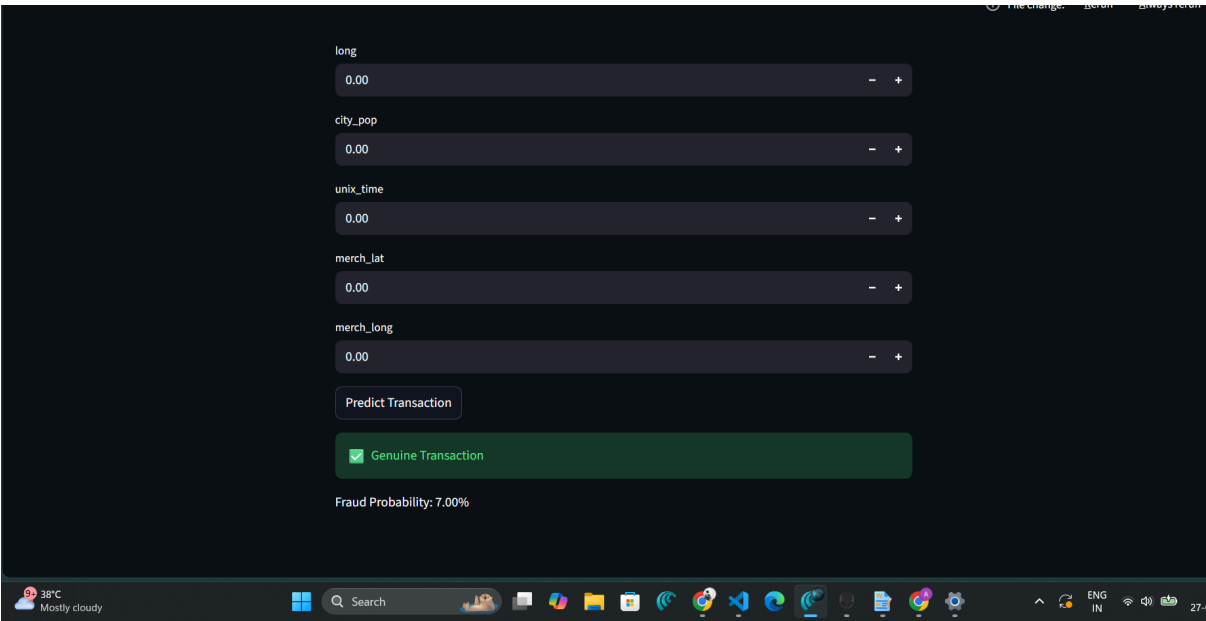
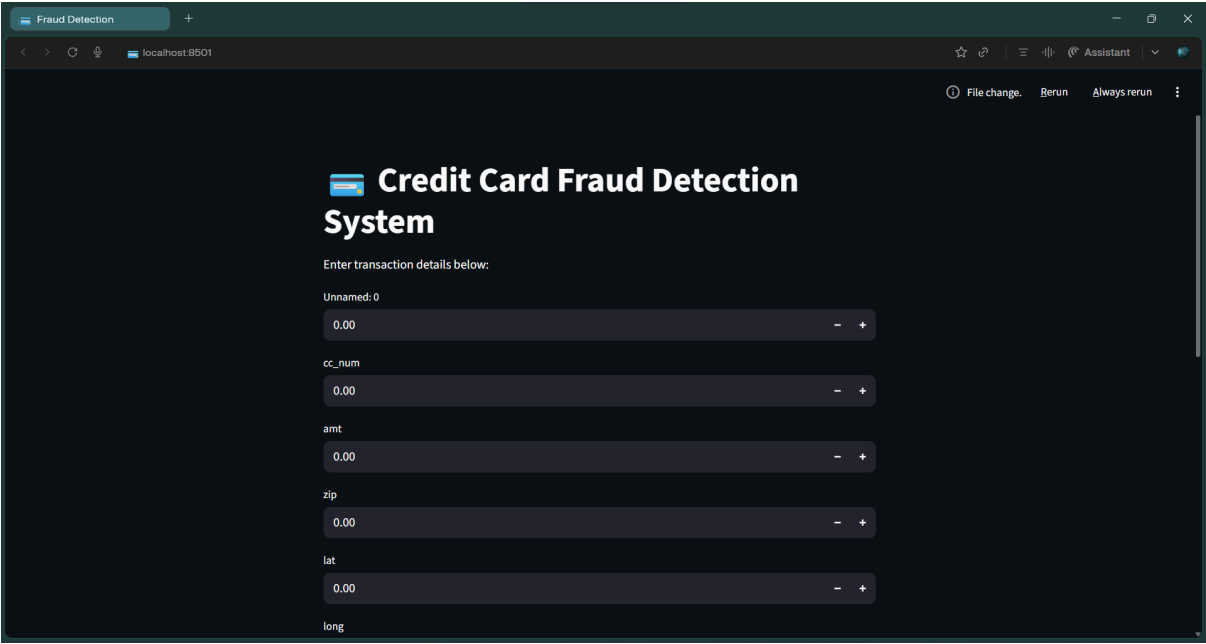
10. Conclusion

The project managed to deploy an entire MLOps pipeline from end to end for building a credit card fraud detection system. Several machine learning models were experimented upon, and Random Forest was found to have performed well with a ROC-AUC score of 0.979 approximately. Preprocessing of data, scaling of features, and using SMOTE made the models perform better due to their high imbalance.

Not only was it important to build machine learning models that performed well but it was equally important to make sure they are reproducible and can be scaled for deployment in production. In order to do so, several MLOps tools were used such as DVC for managing versions of datasets, MLflow for experimenting with models, GitHub actions for CI/CD purposes, Docker for creating containers, and Kubernetes for configuration.

Therefore, not only did the project manage to develop a good fraud detection model but it also succeeded in deploying an entire workflow of building a machine learning system using industry standards.

Screenshots :



```
.git/MERGE_MSG [unix] (15:22 27/05/2026) 1,1 All
```

1,1 All


```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add .
>> git commit -m "Added model files"
>> git push
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git pull origin main --allow-unrelated-histories
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (10/10), 3.08 KiB | 150.00 KiB/s, done.
From https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops
* branch          main          -> FETCH_HEAD
c374422..2634f4c  main          -> origin/main
Merge made by the 'ort' strategy.
.devcontainer/devcontainer.json | 33 ++++++
README.md                      | 34 ++++++
requirements.txt                | 5 +++++
3 files changed, 72 insertions(+)
create mode 100644 .devcontainer/devcontainer.json
create mode 100644 README.md
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```
create mode 100644 README.md
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 595 bytes | 595.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops.git
2634f4c..7a5f75a  main -> main
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add .
>> git commit -m "Fixed model path"
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git push origin main
[main daada06] Fixed model path
1 file changed, 2 insertions(+), 2 deletions(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 311 bytes | 311.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops.git
945ae64..daada06  main -> main
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git rm -r --cached .
rm '.devcontainer/devcontainer.json'
rm '.gitignore'
rm 'ML_Fraud Detection _ Project.ipynb'
rm 'README.md'
rm 'app.py'
rm 'model.py'
rm 'models/DecisionTree.joblib'
rm 'models/LogisticRegression.joblib'
rm 'models/scaler.joblib'
rm 'outputs/DecisionTree.joblib'
rm 'outputs/LogisticRegression.joblib'
rm 'outputs/RandomForest.joblib'
rm 'outputs/roc_curves.png'
rm 'outputs/scaler.joblib'
rm 'plots/model_accuracy_barplot.png'
rm 'plots/roc_curves.png'
rm 'requirements.txt'
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add .
warning: in the working copy of 'ML_Fraud Detection _ Project.ipynb', LF will be replaced by CRLF the next time Git touches it

```

```

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   .gitignore
    new file:   archive/fraudTest.csv
    new file:   archive/fraudTrain.csv
    new file:   data/archive.zip
    modified:   requirements.txt

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>

```

```

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   .gitignore
    deleted:   models/LogisticRegression.joblib
    deleted:   models/scaler.joblib
    deleted:   outputs/DecisionTree.joblib
    deleted:   outputs/LogisticRegression.joblib
    deleted:   outputs/RandomForest.joblib
    deleted:   outputs/scaler.joblib
    modified:   requirements.txt

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git commit -m "Final Streamlit deployment fix"
[main 482c623] Final Streamlit deployment fix
 8 files changed, 6 insertions(+), 12 deletions(-)
 delete mode 100644 models/LogisticRegression.joblib
 delete mode 100644 models/scaler.joblib
 delete mode 100644 outputs/DecisionTree.joblib
 delete mode 100644 outputs/LogisticRegression.joblib
 delete mode 100644 outputs/RandomForest.joblib
 delete mode 100644 outputs/scaler.joblib
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>

```

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git status
deleted:    outputs/RandomForest.joblib
deleted:    outputs/scaler.joblib
modified:   requirements.txt

• (.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git commit -m "Final Streamlit deployment fix"
[main 482c623] Final Streamlit deployment fix
8 files changed, 6 insertions(+), 12 deletions(-)
delete mode 100644 models/LogisticRegression.joblib
delete mode 100644 models/scaler.joblib
delete mode 100644 outputs/DecisionTree.joblib
delete mode 100644 outputs/LogisticRegression.joblib
delete mode 100644 outputs/RandomForest.joblib
delete mode 100644 outputs/scaler.joblib
• (.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 569 bytes | 569.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops.git
cc396ca..482c623 main -> main
○ (.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
Open Website Generate Commit Message Anjaneyulu (22 minutes ago) Ln 11, Col 45 Spaces: 4 UTF-8 CRLF {} Python
```

```
❖ (.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> python model.py
C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops\model.py:40: SyntaxWarning: invalid escape sequence '\D'
ZIP_PATH = "C:\\Users\\rajua\\OneDrive\\Desktop\\project . ML\\archive.zip" # path to uploaded archive (provided i
n conversation)
Looking for CSV files inside the provided zip...
Using data file: /mnt/data/fraudTest.csv

Dataset shape: (555719, 23)

Columns: ['Unnamed: 0', 'trans_date_trans_time', 'cc_num', 'merchant', 'category', 'amt', 'first', 'last', 'gender', '
street', 'city', 'state', 'zip', 'lat', 'long', 'city_pop', 'job', 'dob', 'trans_num', 'unix_time', 'merch_lat', 'merc
h_long', 'is_fraud']

First 5 rows:
   Unnamed: 0  trans_date_trans_time      cc_num  ...  merch_lat  merch_long  is_fraud
0           0   2020-06-21 12:14:25  2291163933867244  ...   33.986391   -81.200714         0
1           1   2020-06-21 12:14:33  3573030041201292  ...   39.450498  -109.960431         0
2           2   2020-06-21 12:14:53  3598215285024754  ...   40.495810   -74.196111         0
3           3   2020-06-21 12:15:15  3591919803438423  ...   28.812398   -80.883061         0
4           4   2020-06-21 12:15:17  3526826139003047  ...   44.959148   -85.884734         0

[5 rows x 23 columns]

Checking for missing values:
Unnamed: 0      0
trans_date_trans_time  0
cc_num          0
merchant        0
category        0
amt             0
first           0
last            0
gender          0
street          0
city            0
state           0
zip             0
lat             0
long            0
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> python model.py
```

```
city          0
state         0
zip           0
lat           0
long          0
city_pop      0
job           0
dob           0
trans_num     0
unix_time     0
dtype: int64
Using target column: is_fraud
```

Class distribution:

```
is_fraud
0      553574
1       2145
```

Name: count, dtype: int64

Dropping non-numeric columns: ['trans_date_trans_time', 'merchant', 'category', 'first', 'last', 'gender', 'street', 'city', 'state', 'job', 'dob', 'trans_num'] (you can preprocess them instead)

Train positive ratio: 0.003859866164314233 Test positive ratio: 0.003859857482185273

Performing SMOTE on training set...

After resampling, class counts: [442859 442859]

Training LogisticRegression...

	precision	recall	f1-score	support
0	0.9989	0.9459	0.9717	110715
1	0.0498	0.7319	0.0933	429
accuracy			0.9451	111144
macro avg	0.5244	0.8389	0.5325	111144
weighted avg	0.9952	0.9451	0.9683	111144

ROC-AUC: 0.8491

Confusion matrix:

```
[[104725  5990]
 [   115   314]]
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> python model.py
```

```
Train positive ratio: 0.003859866164314233 Test positive ratio: 0.003859857482185273
```

```
Performing SMOTE on training set...
```

```
After resampling, class counts: [442859 442859]
```

```
Training LogisticRegression...
```

	precision	recall	f1-score	support
0	0.9989	0.9459	0.9717	110715
1	0.0498	0.7319	0.0933	429
accuracy			0.9451	111144
macro avg	0.5244	0.8389	0.5325	111144
weighted avg	0.9952	0.9451	0.9683	111144

```
ROC-AUC: 0.8491
```

```
Confusion matrix:
```

```
[[104725  5990]  
 [   115   314]]
```

```
Model saved to: outputs\LogisticRegression.joblib
```

```
Training DecisionTree...
```

	precision	recall	f1-score	support
0	0.9988	0.9944	0.9966	110715
1	0.3205	0.6807	0.4358	429
accuracy			0.9932	111144
macro avg	0.6596	0.8375	0.7162	111144
weighted avg	0.9961	0.9932	0.9944	111144

```
ROC-AUC: 0.8375
```

```
Confusion matrix:
```

```
[[110096   619]  
 [   137   292]]
```

```
Model saved to: outputs\DecisionTree.joblib
```

```
Training RandomForest...
```

	precision	recall	f1-score	support
0	0.9991	0.9981	0.9986	110715
1	0.6151	0.7786	0.6872	429
accuracy			0.9973	111144
macro avg	0.8071	0.8883	0.8429	111144
weighted avg	0.9977	0.9973	0.9974	111144

```
ROC-AUC: 0.9875
```

```
Confusion matrix:
```

```
[[110506   209]  
 [    95   334]]
```

```
Model saved to: outputs\RandomForest.joblib
```

```
Saved scalers to output folder.
```

```
ROC plot saved to: outputs\roc_curves.png
```

```
Summary of model AUCs:
```

```
LogisticRegression: AUC = 0.8491
```

```
DecisionTree: AUC = 0.8375
```

```
RandomForest: AUC = 0.9875
```

```
All outputs (models, scalers, plots) are in: outputs
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
>> git add -i models/DecisionTree.joblib
>> git commit -m "Restore scaler model"
>> git push origin main
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add .
>> git commit -m "Fix correct model paths"
>> git push origin main
[main a90059e] Fix correct model paths
5 files changed, 2 insertions(+), 2 deletions(-)
create mode 100644 outputs/DecisionTree.joblib
create mode 100644 outputs/LogisticRegression.joblib
create mode 100644 outputs/RandomForest.joblib
create mode 100644 outputs/scaler.joblib
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 75% (6/8)
```

About Project

This project predicts whether a transaction is fraudulent or genuine using Machine Learning.

Model: Decision Tree

Credit Card Fraud Detection System

Enter transaction details below to predict fraud.

amt

0.00

- +

city_pop

0.00

- +

unix_time

0.00

- +

merch_lat

0.00

- +

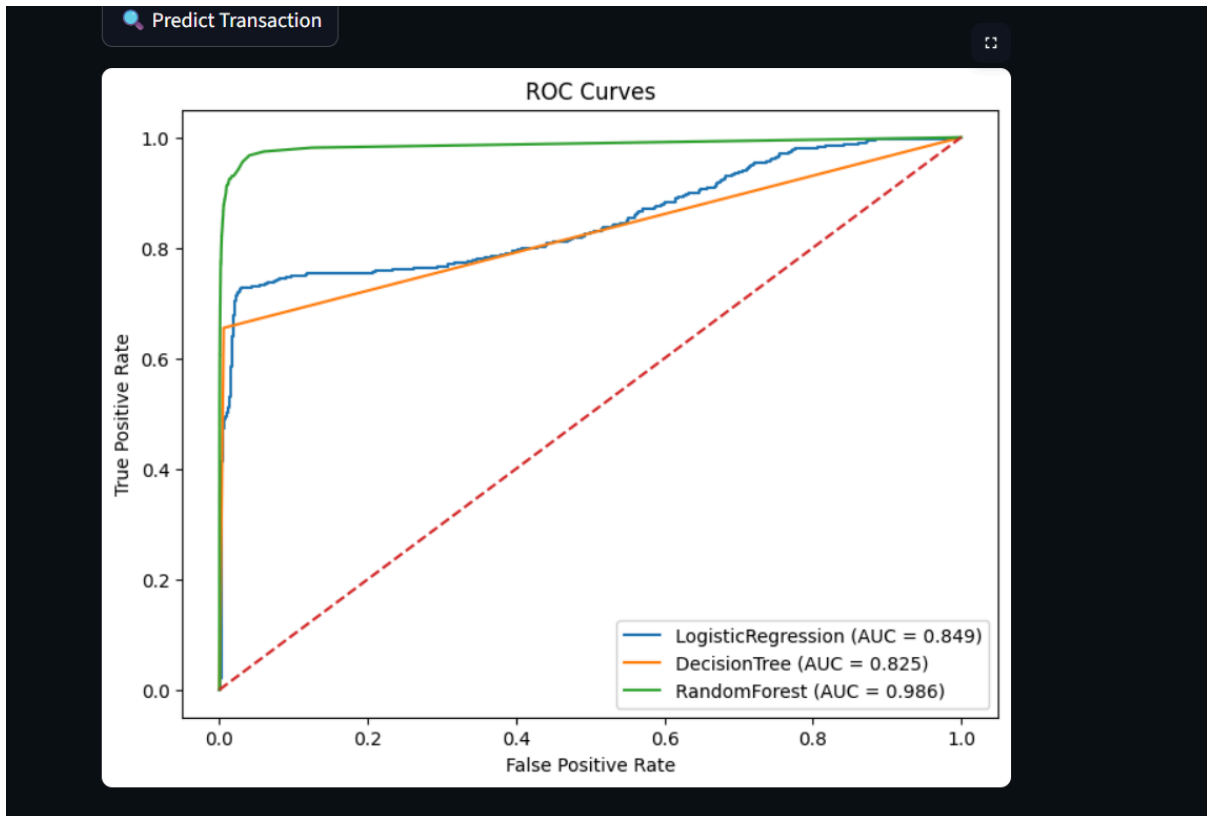
merch_long

0.00

- +

Predict Transaction

Manage app



```
>> git push origin main
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git pull origin main --rebase
remote: Enumerating objects: 23, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (20/20), done.
remote: Total 20 (delta 11), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (20/20), 107.06 KiB | 794.00 KiB/s, done.
From https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops
* branch      main      -> FETCH HEAD
   1d8f484..ccfefe3 main  -> origin/main
Updating 1d8f484..ccfefe3
Fast-forward
 README.md      | 4 ++++
 app_demo.png   | Bin 0 -> 61581 bytes
 plots/app_demo.png | Bin 0 -> 61581 bytes
 plots/app_demo.png -1 | Bin 0 -> 61446 bytes
4 files changed, 4 insertions(+)
create mode 100644 app_demo.png
create mode 100644 plots/app_demo.png
create mode 100644 plots/app_demo.png -1
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

BLACKBOX Agent Open Website Generate Commit Message Anjaneyulu (2 hours ago) Ln 25, Col 2 Spaces: 4 UTF-8 CRLF Pyt

```
FileNotFoundError: [Errno 2] No such file or directory: 'archive/fraud_train.csv'
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> python src/train.py
Dataset Shape: (1296675, 23)
Dropping non-numeric columns:
['trans_date_trans_time', 'merchant', 'category', 'first', 'last', 'gender', 'street', 'city', 'state', 'job', 'dob', 'trans_num']
Applying SMOTE...

Training LogisticRegression...
      precision    recall  f1-score   support

     0       1.00      0.95      0.98    257834
     1       0.09      0.76      0.16      1501

 accuracy
macro avg      0.54      0.86      0.57    259335
weighted avg    0.99      0.95      0.97    259335

ROC-AUC: 0.8351241098542985
Saved model: outputs/LogisticRegression.joblib
```

```
Training LogisticRegression...
      precision    recall  f1-score   support

     0       1.00      0.95      0.98    257834
     1       0.09      0.76      0.16      1501

 accuracy
macro avg      0.54      0.86      0.57    259335
weighted avg    0.99      0.95      0.97    259335

ROC-AUC: 0.8351241098542985
Saved model: outputs/LogisticRegression.joblib

Training DecisionTree...
      precision    recall  f1-score   support

     0       1.00      0.99      0.99    257834
     1       0.31      0.62      0.41      1501

 accuracy
macro avg      0.65      0.80      0.70    259335
weighted avg    0.99      0.99      0.99    259335

ROC-AUC: 0.8044797822367021
Saved model: outputs/DecisionTree.joblib
```

```
Training RandomForest...
      precision    recall  f1-score   support

     0       1.00      1.00      1.00    257834
     1       0.62      0.72      0.67      1501

 accuracy
macro avg      0.81      0.86      0.83    259335
weighted avg    1.00      1.00      1.00    259335

ROC-AUC: 0.9794707993668175
Saved model: outputs/RandomForest.joblib
ROC curve saved: plots/roc_curves.png

Training Complete
All models saved in outputs/
```



```
> dvc initC:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
settings, kombu, dvc-data, asyncssh, aiohttp-retry, typer, scmrepo, dvc-http, celery, gto, dvc-task, dvc
successfully installed aiohttp-retry-2.9.1 amqp-5.3.1 antlr4-python3-runtime-4.9.3 appdirs-1.4.4 asyncssh-2.23.0 atpublic-7.0.
billiard-4.2.4 celery-5.6.3 click-didyoumean-0.3.1 click-plugins-1.1.1.2 click-repl-0.3.0 configobj-5.0.9 dictdiffer-0.9.0 d
skcache-5.6.3 distro-1.9.0 dpath-2.2.0 dulwich-1.2.5 dvc-3.67.1 dvc-data-3.18.3 dvc-http-2.32.0 dvc-objects-5.2.0 dvc-render
.0.2 dvc-studio-client-0.22.0 dvc-task-0.40.2 entrypoints-0.4 filelock-3.29.0 flatten-dict-0.5.0 fluf.lock-9.1.0 fsspec-2026
4.0 funcy-2.0 grandalf-0.8 gto-1.9.0 hydra-core-1.3.2 iterative-telemetry-0.0.10 kombu-5.6.2 markdown-it-py-4.2.0 mdurl-0.1.2
networkx-3.6.1 omegaconf-2.3.0 orjson-3.11.9 pathspec-1.1.1 platformdirs-4.10.0 prompt-toolkit-3.0.52 psutil-7.2.2 pydantic-s
ettings-2.14.1 pydot-4.0.1 pygit2-1.19.2 pygments-2.20.0 pygtrie-2.5.0 rich-15.0.0 ruamel.yaml-0.19.1 scmrepo-3.6.2 semver-3.0
4 setuptools-82.0.1 shellingham-1.5.4 shortuuid-1.0.13 shtab-1.8.0 sqltrie-0.11.2 tabulate-0.10.0 tomlkit-0.15.0 tqdm-4.67.3
typer-0.26.3 tzlocal-5.3.1 vine-5.1.0 voluptuous-0.16.0 zc.lockfile-4.0
initialized DVC repository.
```

You can now commit the changes to git.

```
DVC has enabled anonymous aggregate usage analytics.
Read the analytics documentation (and how to opt-out) here:
<https://dvc.org/doc/user-guide/analytics>
```

What's next?

```
Check out the documentation: <https://dvc.org/doc>
Get help and share ideas: <https://dvc.org/chat>
Star us on GitHub: <https://github.com/treeverse/dvc>
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> dir archive\archive
```

Directory: C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops\archive\archive

Mode	LastWriteTime	Length	Name
-a----	29-05-2026 22:35	150354339	fraudTest.csv
-a----	29-05-2026 22:35	351238196	fraudTrain.csv
-a----	29-05-2026 22:35	8016	Untitled-2.py

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> dvc add archive\archive\fraudTrain.csv
100% Adding... |1/1 [00:01, 1.58s/file]
```

To track the changes with git, run:

```
git add 'archive\archive\.gitignore' 'archive\archive\fraudTrain.csv.dvc'
```

To enable auto staging, run:

```
dvc config core.autostage true
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add archive\archive\.gitignore
```

```
>> git add archive\archive\fraudTrain.csv.dvc
```

```
>>
```

```
>> git commit -m "Track dataset with DVC"
```

```
[main d0563bd] Track dataset with DVC
5 files changed, 12 insertions(+)
create mode 100644 .dvc/.gitignore
create mode 100644 .dvc/config
create mode 100644 .dvcignore
create mode 100644 archive/archive/.gitignore
create mode 100644 archive/archive/fraudTrain.csv.dvc
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> python src/train.py
Dataset Shape: (1296675, 23)
Dropping non-numeric columns:
['trans_date_trans_time', 'merchant', 'category', 'first', 'last', 'gender', 'street', 'city', 'state', 'job', 'dob', 'trans_num']
Applying SMOTE...

Training LogisticRegression...
      precision    recall  f1-score   support

     0       1.00      0.95      0.98    257834
     1       0.09      0.76      0.16      1501

 accuracy
macro avg      0.54      0.86      0.57    259335
weighted avg    0.99      0.95      0.97    259335

ROC-AUC: 0.8351241098542985
Saved model: outputs/LogisticRegression.joblib

Training DecisionTree...
      precision    recall  f1-score   support

     0       1.00      0.99      0.99    257834
     1       0.31      0.62      0.41      1501

 accuracy
macro avg      0.65      0.80      0.70    259335
weighted avg    0.99      0.99      0.99    259335

ROC-AUC: 0.8044797822367021
Saved model: outputs/DecisionTree.joblib
```

```
Training DecisionTree...
      precision    recall  f1-score   support

     0       1.00      0.99      0.99    257834
     1       0.31      0.62      0.41      1501

 accuracy
macro avg      0.65      0.80      0.70    259335
weighted avg    0.99      0.99      0.99    259335

ROC-AUC: 0.8044797822367021
Saved model: outputs/DecisionTree.joblib

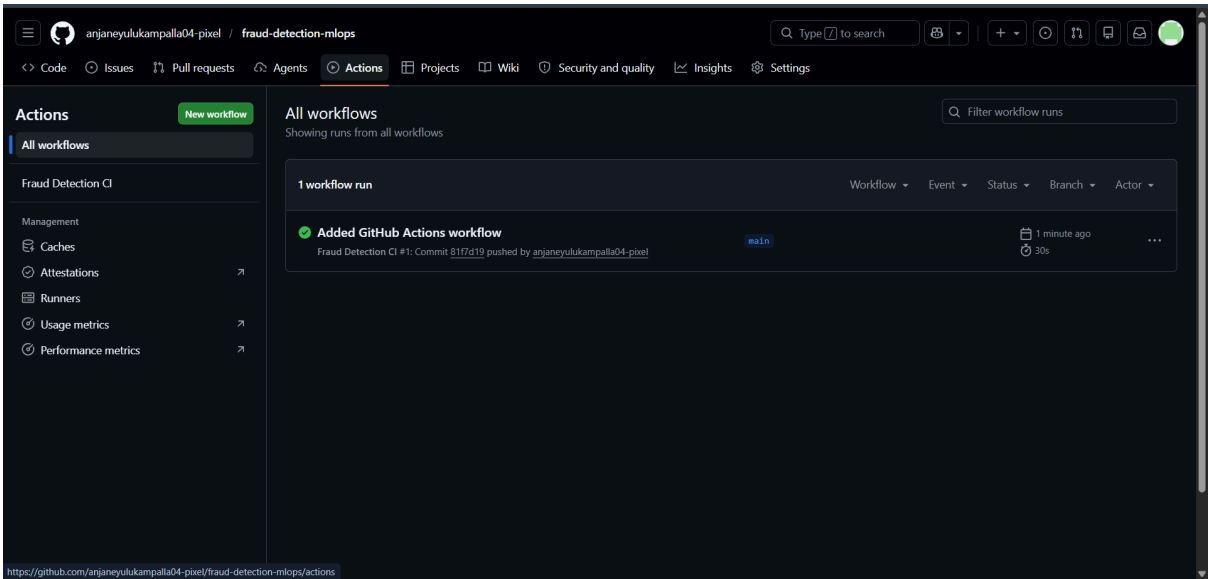
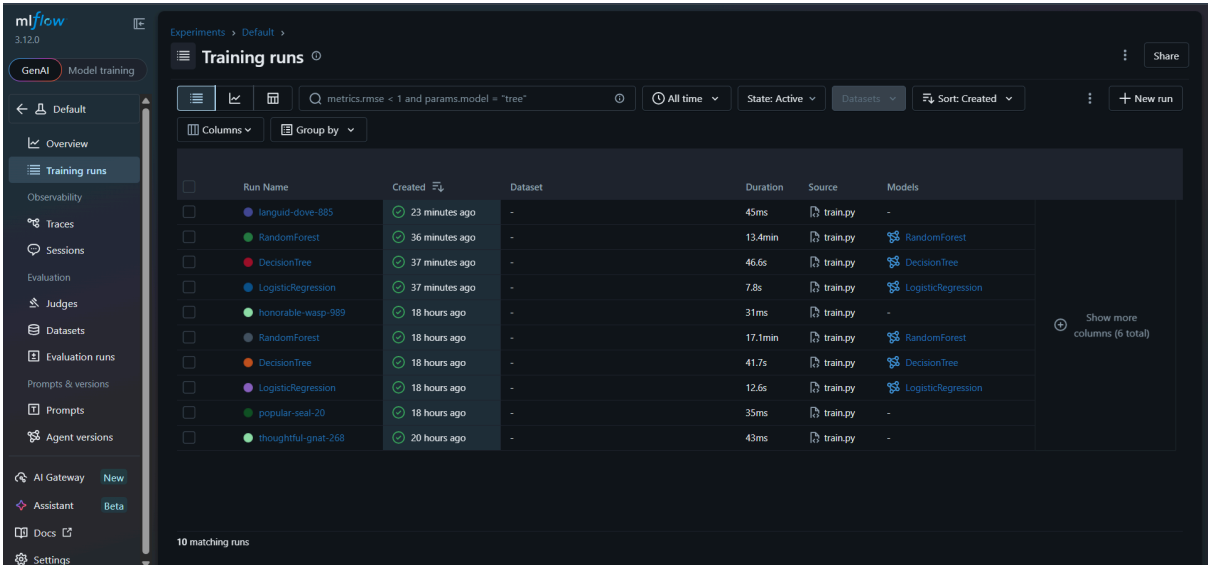
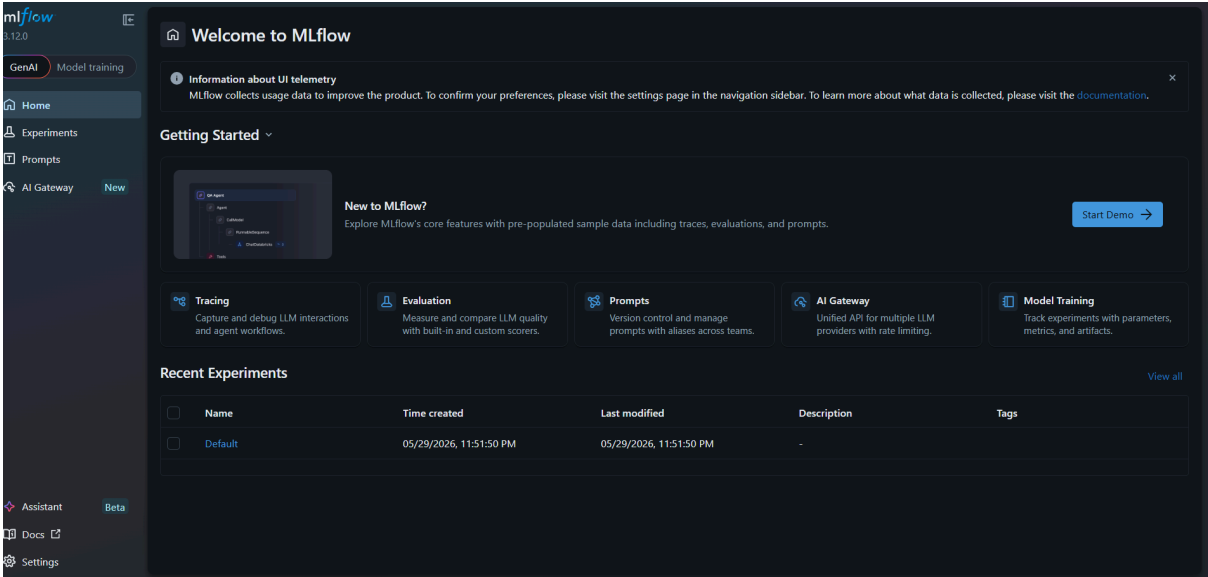
Training RandomForest...
      precision    recall  f1-score   support

     0       1.00      1.00      1.00    257834
     1       0.62      0.72      0.67      1501

 accuracy
macro avg      0.81      0.86      0.83    259335
weighted avg    1.00      1.00      1.00    259335

ROC-AUC: 0.9794707993668175
Saved model: outputs/RandomForest.joblib
ROC curve saved: plots/roc_curves.png

Training Complete
All models saved in outputs/
2026/05/29 23:51:49 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/05/29 23:51:49 INFO mlflow.store.db.utils: Updating database tables
```

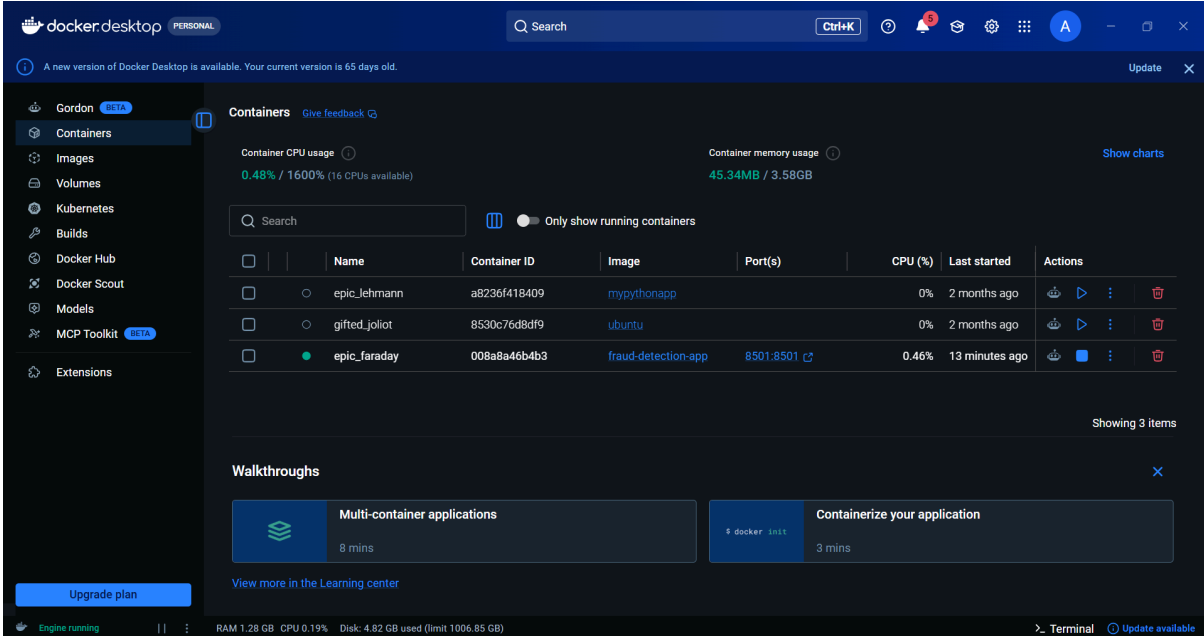


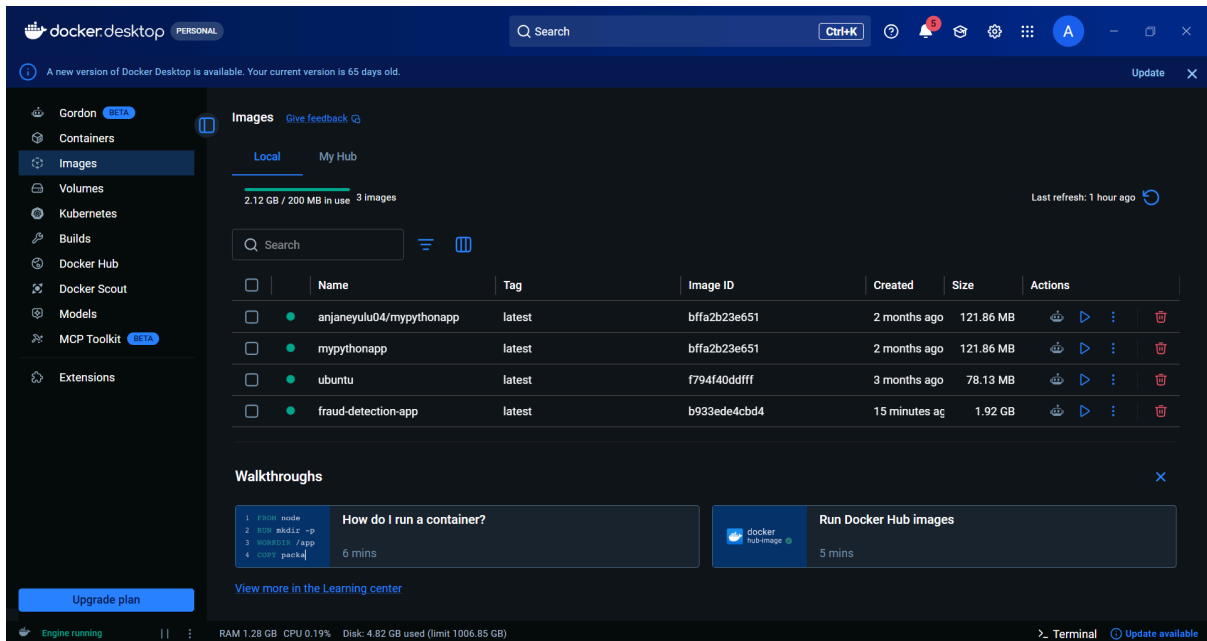
```
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> docker build -t fraud-detection-app .
=> => extracting sha256:797d495f2c68eb4664df918d39078ca04612d7ef47c57b5c784f7f6eadd42bf5 1.0s
=> => extracting sha256:45006ceeeea9e2ca59a046b6f4ac9a212e4b27b3d8d6e7d348b03f0aaccdac99 0.0s
=> [internal] load build context 258.3s
=> => transferring context: 1.17GB 258.0s
=> [2/5] WORKDIR /app 0.1s
=> [3/5] COPY requirements.txt . 0.2s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt 84.1s
=> [5/5] COPY . . 20.7s
=> exporting to image 13.2s
=> => exporting layers 13.1s
=> => writing image sha256:b933ede4cbd4b42b1ebadb3712331857e2f3b316ad6110a08c8aea73f9bd7261 0.0s
=> => naming to docker.io/library/fraud-detection-app 0.0s
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> docker images

INFO → U In Use

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
anjaneyulu04/mypythonapp:latest      bffa2b23e651                     122MB       0B            U
fraud-detection-app:latest           b933ede4cbd4                     1.92GB      0B            U
mypythonapp:latest                   bffa2b23e651                     122MB       0B            U
ubuntu:latest                        f794f40ddfff                     78.1MB      0B            U

(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```





```
PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> & c:/Users/rajua/OneDrive/Desktop/fraud-detection-mlops/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops> git add .
>> git commit -m "Added Kubernetes deployment files"
>> git push origin main
[main c3e26ef] Added Kubernetes deployment files
4 files changed, 62 insertions(+)
create mode 100644 .dockerignore
create mode 100644 Dockerfile
create mode 100644 k8s/deployment.yaml
create mode 100644 k8s/service.yaml
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1017 bytes | 1017.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/anjaneyulukampalla04-pixel/fraud-detection-mlops.git
81f7d19..c3e26ef main -> main
(.venv) PS C:\Users\rajua\OneDrive\Desktop\fraud-detection-mlops>
```

anjaneyulukampalla04-pixel / fraud-detection-mlops

Code

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Go to file

Add file

Code

Anjaneyulu

Added Kubernetes deployment files

✓

c3e26ef · 1 minute ago

30 Commits

devcontainer	Added Dev Container Folder	3 days ago
dvc	Track dataset with DVC	yesterday
.github/workflows	Added GitHub Actions workflow	1 hour ago
k8s	Added Kubernetes deployment files	1 minute ago
models	Final Streamlit deployment fix	3 days ago
plots	Remove large files and add MLflow integration	1 hour ago
src	Remove large files and add MLflow integration	1 hour ago
.dockerignore	Added Kubernetes deployment files	1 minute ago
.dvcignore	Track dataset with DVC	yesterday
.gitignore	Remove large files and add MLflow integration	1 hour ago
Dockerfile	Added Kubernetes deployment files	1 minute ago

About

Credit Card Fraud Detection Machine Learning Project

Readme

Activity

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.github/workflows

Added GitHub Actions workflow

1 hour ago

k8s

Added Kubernetes deployment files

1 minute ago

models

Final Streamlit deployment fix

3 days ago

plots

Remove large files and add MLflow integration

1 hour ago

src

Remove large files and add MLflow integration

1 hour ago

.dockerignore

Added Kubernetes deployment files

1 minute ago

.dvcignore

Track dataset with DVC

yesterday

.gitignore

[Remove large files and add MLflow integration](#)

1 hour ago

Dockerfile

Added Kubernetes deployment files

1 minute ago

ML_Fraud Detection_ Project.ipynb

Initial clean commit

4 days ago

README.md

Update README.md

2 days ago

app.py

Final working fraud detection project

2 days ago

app_demo.png

Rename plots/[35C8F586-5E23-43A1-BD99-8AA6265D3B5D]...

2 days ago

model.py

Initial clean commit

4 days ago

requirements.txt

Final Streamlit deployment fix

3 days ago

README

0 watching

0 forks

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anjaneyulukampalla04-pixel

Languages

Jupyter Notebook 94.9%

Python 5.0%

Dockerfile 0.1%

README

Credit Card Fraud Detection System

Project Overview

This project is a Machine Learning based Credit Card Fraud Detection System developed using Python, Scikit-learn, and Streamlit.

The system predicts whether a transaction is genuine or fraudulent based on transaction-related features.

Features

- Fraud Detection
- Machine Learning Models
- Streamlit Web App
- ROC Curve Evaluation

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- GitHub

Models Used

1. Logistic Regression
2. Decision Tree
3. Random Forest

Best Model

Random Forest achieved highest performance.

Run Project

```
pip install -r requirements.txt
streamlit run app.py
```



```
## Application Demo
```

```
!["Demo"](plots/app_demo.png)
```